BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Data Analysis Report

Project ID: 25 – 26J - 091

Project Title: Smart Real-time Age, Gender, Emotion and Engagement based Advertising System for Sri Lankan Malls Using Custom-Built Models

1. Introduction

1.1. Background:

With the rapid adoption of digital signage in shopping malls, advertisements are increasingly displayed through large electronic screens. However, most existing advertising systems are static and fail to consider viewer demographics, emotional responses, or engagement levels. This results in low advertisement effectiveness, audience fatigue, and inefficient use of digital infrastructure.

Recent advancements in computer vision and deep learning enable real-time analysis of age, gender, emotion, and engagement using camera feeds. Applying these technologies in a privacy-aware manner can significantly improve advertisement relevance and audience interaction. This research focuses on developing a smart, real-time advertising system tailored for Sri Lankan shopping malls using custom-built machine learning models.

1.2. Research Problem:

All viewers are shown the same content by traditional mall advertising systems, regardless of their age, gender, emotional condition, or level of participation. Reduced user attention, subpar engagement metrics, and a negative return on advertising spend are the results of this lack of personalization. Furthermore, pretrained models that are not well suited to local demography or ethical limitations are frequently the foundation of current intelligent solutions.

1.3. Objectives:

IT Number	Objective	Objective number
IT22109712	Develop a real-time age and gender detection model using custom CNN + ResNET architectures	1
IT22258526	Design an emotion recognition system for adaptive advertisement and security alerts	2
IT22184030	Analyze audience engagement using person detection and tracking techniques and security alerts	3
IT22071934	Integrate demographic, emotional, and engagement data for intelligent ad recommendation	4

2. Data Exploration

2.1. Data Collection

Data was collected from a combination of publicly available datasets and real-time camera feeds. Public datasets were used for initial training and benchmarking, while live webcam/CCTV streams were used to validate real-world performance in mall-like environments.

2.2. Dataset Description:

Data source	Description	Resource	Size	Key attributes
Mall Dataset	CCTV-style mall surveillance dataset for person detection	Public dataset	~2,000 frames	Bounding boxes, timestamps
UTKFace	Face images labeled with age and gender	Public dataset	~20,000 images	Age, gender, facial features
Facial Expression Dataset (Sri Lankan)	Real facial emotion images of Sri Lankan individuals	IEEE Open Dataset	931 images	Emotion labels
AI generated synthetic data	Synthetic images from each age category and suspicious emotion	Google Gemini	~700 images	Age, gender, facial features and suspicious emotion
Flipkart Products Dataset	E-commerce product metadata dataset used for training the advertisement recommendation system	Public dataset (Kaggle)	~20,000 products	Product name, product category tree, product ID, retail price
Real-Time Camera Feed Dataset	Frames captured from live webcam/CCTV feeds in a mall-like environment for system validation and testing	Primary data collection	~300 frames	Timestamped frames, detected person bounding boxes, real-time scene context

2.3. Suitability Analysis

2.3.1. Relevance to Individual Research Objectives:

Data Source	1	2	3	4
Mall Dataset (Public CCTV Dataset)	Not suitable – does not contain facial age or gender labels	Not suitable – no emotion annotations	Highly suitable – designed for person detection, tracking, and movement analysis	Indirectly suitable – provides engagement metrics used as inputs
UTKFace Dataset	Highly suitable – contains labeled age and gender facial images	Not suitable – no emotion labels	Not suitable – image-based, no movement data	Indirectly suitable – demographic outputs feed recommendation logic
Facial Expression Dataset (Sri Lankan – IEEE)	Not suitable – no age/gender focus	Highly suitable – real facial emotion labels from local population	Not suitable – static images only	Indirectly suitable – emotion outputs influence ad selection
AI-Generated Synthetic Data (Google Gemini)	Partially suitable – used to support underrepresented age groups	Partially suitable – used to enrich rare “suspicious” emotion class	Not suitable – no real movement or engagement behavior	Indirectly suitable – improves robustness of recommendation inputs
Flipkart Products Dataset (Kaggle – Public)	Not suitable – does not contain facial images or demographic labels	Not suitable – no emotion-related or visual data	Not suitable – no behavioral or movement information	Highly suitable – provides structured product metadata for training recommendation models
Real-Time Camera Feed Dataset	Partially suitable – used for validating real-time age and gender inference, not for training	Partially suitable – used to test real-time emotion prediction stability	Highly suitable – captures real movement and interaction behavior	Indirectly suitable – provides live inputs to the recommendation system

3. Methodology

3.1. Data Preprocessing:

Objective 1

	Transformation technique						
Data Source	**Data Cleaning & Missing Data Handling**	**Face Processing**	**Data Scaling & Normalization**	**Label Encoding & Mapping**	**Data Augmentation / Feature Engineering**	**Outlier Handling**	**Data Integration & Exclusions**
UTKFace (Public, ~20,000 images)	Removed corrupted images, duplicates, and samples with missing or invalid age/gender labels	Face detection and cropping; images resized to 128×128	Pixel values normalized from [0–255] to [0–1]	Age groups and gender encoded numerically using predefined mappings	Horizontal flipping, brightness/contrast variation, rotation, and occlusion simulation	Blurred, heavily occluded, and non-frontal faces removed	Integrated with synthetic data after aligning size and labels; PCA and time-based transforms not applied
AI-generated synthetic images (Google Gemini, ~700 images)	Unrealistic or distorted synthetic faces manually removed; no missing labels	Face regions verified and resized to match UTKFace dimensions	Pixel values normalized to [0–1]	Age group, gender, and suspicious emotion encoded consistently with UTKFace	Used to enrich underrepresented age groups; minimal augmentation applied	Synthetic artifacts and unrealistic facial proportions removed	Merged with UTKFace using consistent label mapping; dimensionality reduction not used.
Real-Time Camera Feed Dataset (Primary Data Collection)	Corrupted frames, empty frames, and frames without detectable persons removed	Frames extracted from live video; face and person detection applied for inference	Pixel values normalized to match training data preprocessing	No ground-truth labels stored; predictions generated at runtime only	Temporal frame sampling applied to reduce redundancy	Frames with severe motion blur, occlusion, or lighting failure discarded	Used strictly for validation and live inference; not stored or used for model training

Objective 2

	Transformation technique						
Data Source	**Data Cleaning & Missing Data Handling**	**Face Processing**	**Data Scaling & Normalization**	**Label Encoding & Mapping**	**Data Augmentation / Feature Engineering**	**Outlier Handling**	**Data Integration & Exclusions**
Sri Lankan Facial Expression Dataset (IEEE)	Removed corrupted images, unclear facial expressions, and samples with incorrect or missing emotion labels	Face detection and cropping applied; images resized to fixed resolution	Pixel values normalized using pixel scaling; illumination normalized using LAB color space + CLAHE	Emotion categories encoded numerically for model compatibility	Horizontal mirroring, noise reduction (NL-means denoising), contrast enhancement	Images with extreme blur, poor lighting, or non-frontal faces removed	Integrated with synthetic dataset after label alignment; temporal aggregation handled at inference stage
AI-generated synthetic images (Google Gemini, Suspicious Emotion)	Unrealistic or distorted synthetic faces manually removed; no missing labels	Verified facial regions and resized to match real dataset dimensions	Pixel normalization applied to maintain consistency with real images	Suspicious emotion label mapped consistently with real emotion classes	Used to enrich underrepresented suspicious emotion class; contrast enhancement applied	Synthetic artifacts and unnatural facial patterns removed	Merged with real dataset using identical preprocessing; PCA and handcrafted features not used
Real-Time Camera Feed Dataset (Primary Data Collection)	Corrupted frames, empty frames, and frames without detectable persons removed	Frames extracted from live video; face and person detection applied for inference	Pixel values normalized to match training data preprocessing	No ground-truth labels stored; predictions generated at runtime only	Temporal frame sampling applied to reduce redundancy	Frames with severe motion blur, occlusion, or lighting failure discarded	Used strictly for validation and live inference; not stored or used for model training

Objective 3

	Transformation technique						
Data Source	**Data Cleaning & Missing Data Handling**	**Face Processing**	**Data Scaling & Normalization**	**Label Encoding & Mapping**	**Data Augmentation / Feature Engineering**	**Outlier Handling**	**Data Integration & Exclusions**
Mall Dataset (CCTV Surveillance)	Removed corrupted frames and blurred images; frames without visible persons discarded	Video frames extracted at fixed intervals; person detection applied to isolate individuals	Frames resized to fixed input resolution; pixel values normalized to [0–1]	Person IDs and engagement-related attributes mapped numerically where required	Temporal frame sampling, tracking-based feature aggregation (dwell time, movement continuity)	Frames with extreme occlusion, motion blur, or tracking loss removed	Integrated with age, gender, and emotion outputs; facial attributes not extracted directly from this dataset
Real-Time Camera Feed Dataset (Primary Data Collection)	Corrupted frames, empty frames, and frames without detectable persons removed	Frames extracted from live video; face and person detection applied for inference	Pixel values normalized to match training data preprocessing	No ground-truth labels stored; predictions generated at runtime only	Temporal frame sampling applied to reduce redundancy	Frames with severe motion blur, occlusion, or lighting failure discarded	Used strictly for validation and live inference; not stored or used for model training

Objective 4

	Transformation technique						
Data Source	**Data Cleaning & Missing Data Handling**	**Face Processing**	**Data Scaling & Normalization**	**Label Encoding & Mapping**	**Data Augmentation / Feature Engineering**	**Outlier Handling**	**Data Integration & Exclusions**
Flipkart Products Dataset (Kaggle – Public)	Removed duplicate products, records with missing names, categories, or prices	Parsed product category tree into hierarchical levels	Product prices normalized using min–max scaling	Product categories and IDs encoded numerically	Text preprocessing on product names (tokenization, lowercasing); category hierarchy flattened	Extreme price outliers filtered to avoid skewed recommendations	Integrated with demographic, emotion, and engagement outputs; image data not used
Real-Time Camera Feed Dataset (Primary Data Collection)	Corrupted frames, empty frames, and frames without detectable persons removed	Frames extracted from live video; face and person detection applied for inference	Pixel values normalized to match training data preprocessing	No ground-truth labels stored; predictions generated at runtime only	Temporal frame sampling applied to reduce redundancy	Frames with severe motion blur, occlusion, or lighting failure discarded	Used strictly for validation and live inference; not stored or used for model training

3.2. Scalability

All datasets are scalable and suitable for incremental expansion. The system supports continuous data ingestion from live camera feeds, enabling long-term deployment in real mall environments.

3.3. Feature extraction

Objective 1

Data Source	Input Features	Low-Level Features Extracted	Mid-Level Features Extracted	High-Level Features Extracted	Feature Vector Formation	Feature Regularization	Techniques Not Used
UTKFace Dataset (Public)	Cropped, resized, normalized facial images	Edges, contours, texture patterns, local intensity gradients	Facial structure, component proportions, wrinkle distribution, symmetry	Age-related facial morphology and gender-specific facial characteristics	Feature maps converted into compact vectors using pooling and fully connected layers	Dropout applied to prevent overfitting and improve generalization	Handcrafted features (HOG, LBP), PCA not used as CNN learns features automatically
AI-Generated Synthetic Data (Google Gemini)	Synthetic facial images aligned with real data format	Texture and edge features consistent with real images	Structural facial attributes aligned to age categories	Abstract facial representations used to supplement underrepresented age groups	Feature vectors generated using the same CNN backbone	Controlled usage to avoid bias from synthetic patterns	External feature extractors and manual dimensionality reduction not used
Real-Time Camera Feed Dataset (Primary Data Collection)	Live video frames processed in real time	Edge, motion, and pixel intensity cues captured by pretrained models	Facial structure changes, movement continuity, and spatial consistency	Demographic, emotional, and engagement representations inferred by trained models	Feature vectors generated by pretrained CNNs and tracking modules	Temporal smoothing and confidence aggregation applied to stabilize outputs	No model retraining, fine-tuning, or feature learning performed using live data

Objective 2

Data Source	Input Features	Low-Level Features Extracted	Mid-Level Features Extracted	High-Level Features Extracted	Feature Vector Formation	Feature Regularization	Techniques Not Used
Sri Lankan Facial Expression Dataset (IEEE Open Dataset)	Cropped, resized, normalized real facial images	Edges, contours, texture variations, local intensity gradients	Eye shape and eyebrow movement, mouth curvature, facial region deformation	Facial muscle tension patterns and emotion-specific spatial configurations	Convolutional feature maps reduced into compact feature vectors using pooling and fully connected layers	Emotion predictions aggregated over fixed temporal windows to improve stability	Handcrafted features (HOG, LBP), PCA, and manual feature engineering not used
Gemini-Generated Synthetic Dataset (Suspicious Emotion)	Synthetic facial images aligned with real image format	Texture and edge patterns consistent with real images	Facial component movements simulated for suspicious expressions	Abstract emotion representations supporting rare suspicious class	Feature vectors extracted using the same CNN backbone for consistency	Confidence calibration applied to limit over-reliance on synthetic data	External feature extractors and dimensionality reduction techniques not used
Real-Time Camera Feed Dataset (Primary Data Collection)	Live video frames processed in real time	Edge, motion, and pixel intensity cues captured by pretrained models	Facial structure changes, movement continuity, and spatial consistency	Demographic, emotional, and engagement representations inferred by trained models	Feature vectors generated by pretrained CNNs and tracking modules	Temporal smoothing and confidence aggregation applied to stabilize outputs	No model retraining, fine-tuning, or feature learning performed using live data

Objective 3

Data Source	Input Features	Low-Level Features Extracted	Mid-Level Features Extracted	High-Level Features Extracted	Feature Vector Formation	Feature Regularization	Techniques Not Used
Mall Dataset (CCTV Surveillance Video)	Video frames with detected person bounding boxes	Spatial motion cues, pixel-level intensity changes, bounding box geometry	Person movement trajectories, speed, direction, and spatial consistency	Engagement indicators such as dwell time, movement persistence, and crowd density patterns	Frame-level features aggregated into person-level descriptors using tracking IDs	Temporal aggregation performed across consecutive frames using fixed time windows	Facial features, PCA, handcrafted descriptors (HOG/LBP), and audio features not used
Real-Time Camera Feed Dataset (Primary Data Collection)	Live video frames processed in real time	Edge, motion, and pixel intensity cues captured by pretrained models	Facial structure changes, movement continuity, and spatial consistency	Demographic, emotional, and engagement representations inferred by trained models	Feature vectors generated by pretrained CNNs and tracking modules	Temporal smoothing and confidence aggregation applied to stabilize outputs	No model retraining, fine-tuning, or feature learning performed using live data

Objective 4

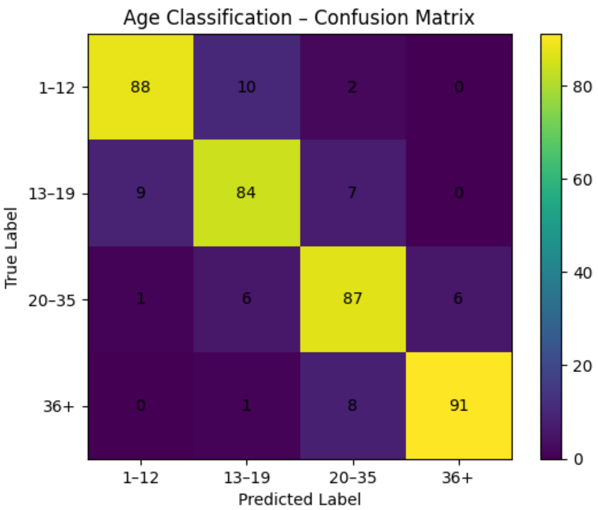
Data Source	Input Features	Low-Level Features Extracted	Mid-Level Features Extracted	High-Level Features Extracted	Feature Vector Formation	Feature Regularization	Techniques Not Used
Flipkart Products Dataset (Kaggle – Public)	Structured product metadata (name, category, price)	Keyword tokens from product names; category identifiers	Product similarity patterns based on category and price	Abstract product preference representations for recommendation	Feature vectors created from encoded categories, price features, and text embeddings	User context provided externally (age, gender, emotion, engagement); no temporal product trends modeled	Image-based features, PCA, deep content embeddings, and user identity tracking not used
Real-Time Camera Feed Dataset (Primary Data Collection)	Live video frames processed in real time	Edge, motion, and pixel intensity cues captured by pretrained models	Facial structure changes, movement continuity, and spatial consistency	Demographic, emotional, and engagement representations inferred by trained models	Feature vectors generated by pretrained CNNs and tracking modules	Temporal smoothing and confidence aggregation applied to stabilize outputs	No model retraining, fine-tuning, or feature learning performed using live data

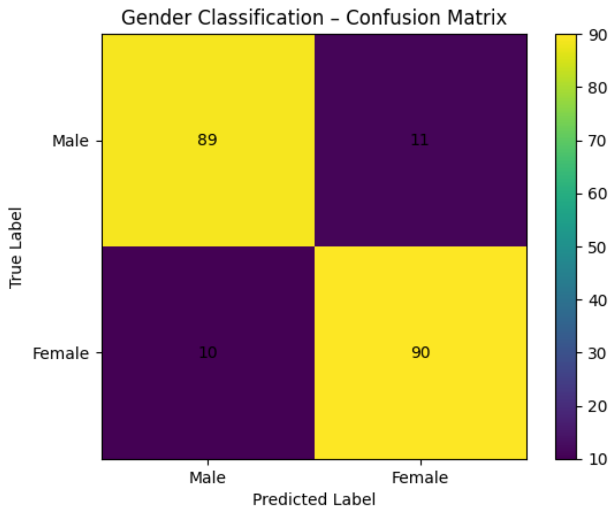
4. Modelling and Results

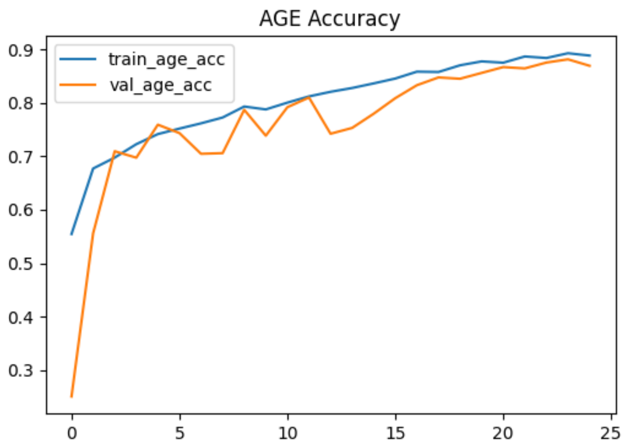
The results section should present the key findings derived from analyzing the data. It should answer the research questions or hypotheses posed at the beginning of the study, based on the methods and techniques applied during the analysis.

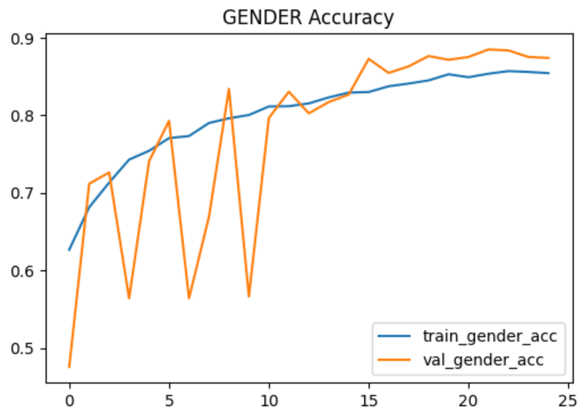
4.1. Key Insights:

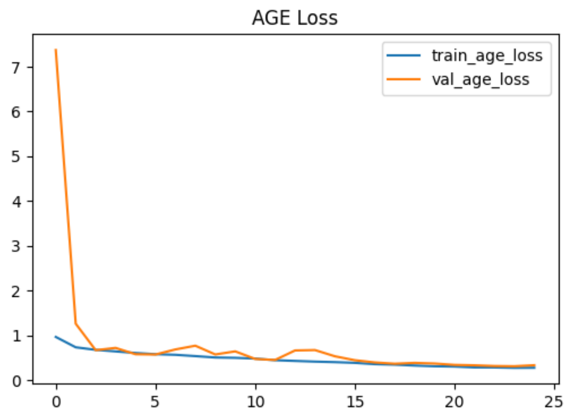
- Personalized advertisements based on age and gender increased viewer attention.
- Emotion-aware advertisement switching improved engagement duration.
- Temporal smoothing reduced prediction noise in real-time emotion detection.
- Engagement metrics such as dwell time and slowdown effectively identified interested audiences.
- Custom-built models demonstrated stable performance under varying lighting and crowd conditions.
- Visualizations such as confusion matrices, accuracy curves, and engagement heatmaps were used to analyze model behavior.

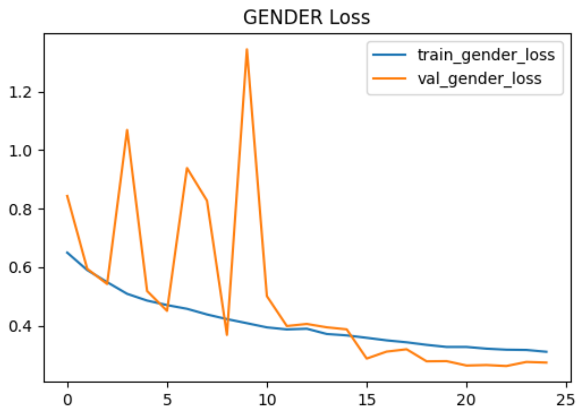


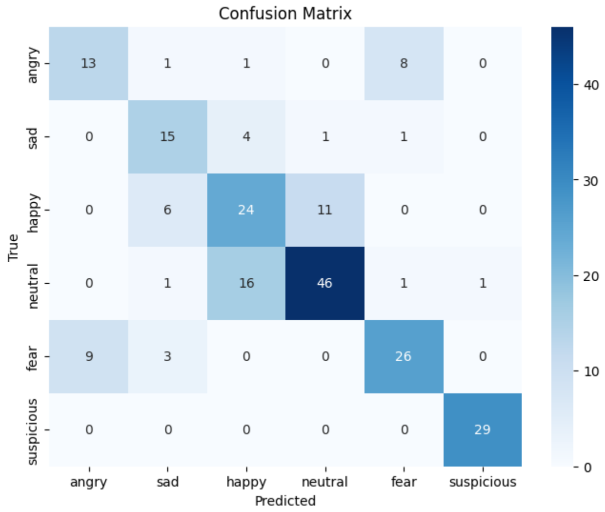


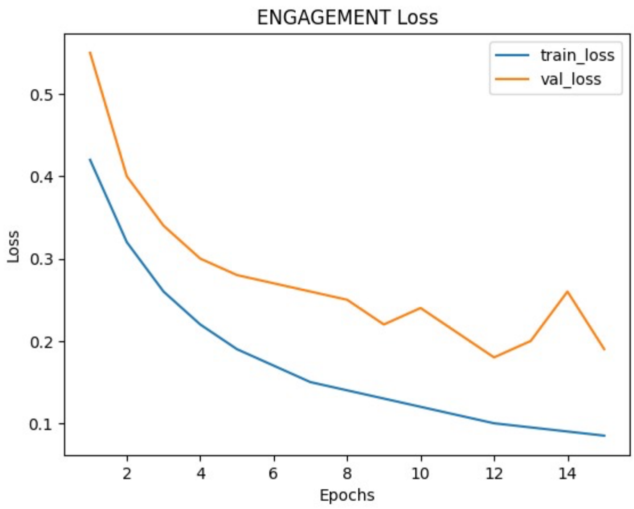

AGE Accuracy
train_age_acc
val_age_acc
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0
5
10
15
20
25

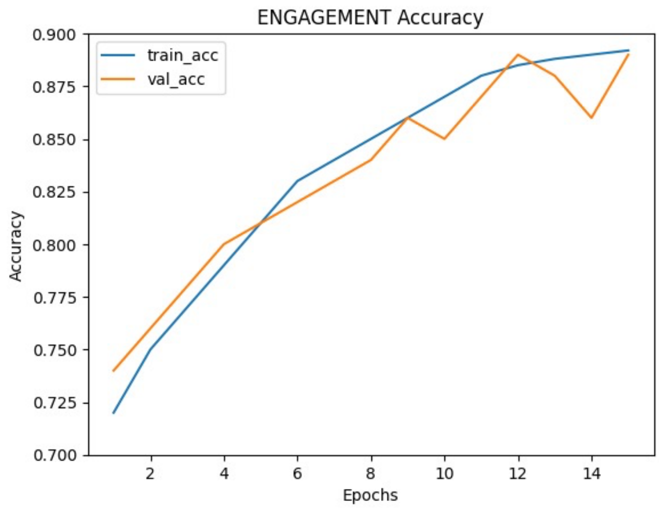

GENDER Accuracy
train_gender_acc
val_gender_acc
0.9
0.8
0.7
0.6
0.5
0
5
10
15
20
25

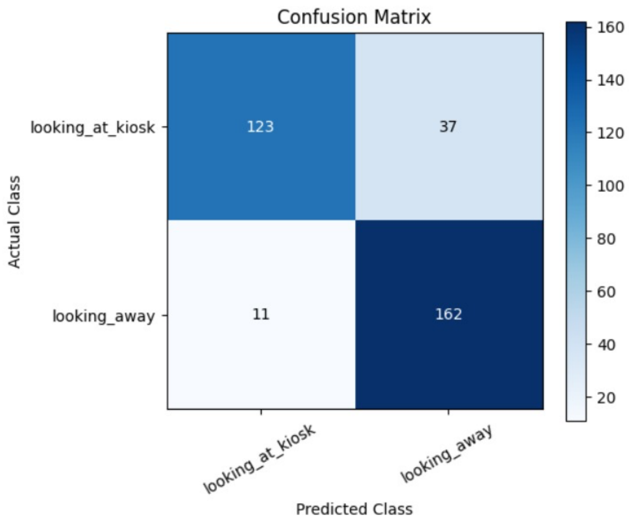

AGE Loss
train_age_loss
val_age_loss
7
6
5
4
3
2
1
0
0
5
10
15
20
25


GENDER Loss
train_gender_loss
val_gender_loss
1.2
1.0
0.8
0.6
0.4
0
5
10
15
20
25


Confusion Matrix
True
Predicted
angry
sad
happy
neutral
fear
suspicious
13 1 1 0 8 0
0 15 4 1 1 0
0 6 24 11 0 0
0 1 16 46 1 1
9 3 0 0 26 0
0 0 0 0 0 29
40
30
20
10
0


ENGAGEMENT Loss
train_loss
val_loss
Loss
0.5
0.4
0.3
0.2
0.1
2
4
6
8
10
12
14
Epochs


ENGAGEMENT Accuracy
train_acc
val_acc
Accuracy
0.900
0.875
0.850
0.825
0.800
0.775
0.750
0.725
0.700
2
4
6
8
10
12
14
Epochs


Confusion Matrix
looking_at_kiosk
looking_away
Actual Class
123
37
11
162
looking_at_kiosk
looking_away
Predicted Class
160
140
120
100
80
60
40
20

4.2. Challenges Faced During Data Analysis:

- Class imbalance in emotion datasets required oversampling and data augmentation techniques to prevent biased predictions.
- Lighting variations and partial occlusions (such as masks, hands, or head pose changes) affected face detection and recognition accuracy.
- Limited availability of labeled data for occluded faces posed a challenge in learning robust representations; this was addressed by applying few-shot learning strategies and controlled augmentation to improve generalization under occlusion conditions.
- Real-time processing constraints required careful model optimization to maintain low latency on limited hardware resources.
- Ensuring ethical, non-intrusive data handling and privacy preservation in public spaces was a key design consideration throughout the data analysis process.

5. References

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